

A Machine Learning Based Approach for the Prediction of Road Traffic Flow on Urbanised Arterial Roads

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Abstract—Urbanised arterial roads connect geographically important areas and are used for commuting and the movement of goods. Prediction of traffic flow on these roads is vital to aid in the mitigation of congestion. However, there is currently a lack of research in this area.

In this work we have applied machine learning models to a real dataset for the prediction of road traffic congestion on urbanised arterial road. A comparative analysis was conducted on each machine learning model, examining the prediction accuracy and time-horizon sensitivity. Furthermore, we examined different input parameter settings (various classes of vehicles such as motorcycles, cars, vans, rigid goods lorries, articulated heavy goods vehicles (HGVs), and buses) to investigate how heterogeneous traffic flow can affect prediction.

The experimental results show that the Artificial Neural Network Model outperforms other models at predicting short-term traffic flow on an urbanised arterial road based on the standard performance indicator: Root Mean Squared Error (RMSE). Additionally, it is found that different classes of vehicles can aid the improvement prediction.

Index Terms—Machine Learning, Short-term Traffic Prediction, K Nearest Neighbours (KNN), Support Vector Regression (SVR), Support Vector Machine (SVM) and Artificial Neural Networks (ANN).

I. INTRODUCTION

Traffic congestion is a major issue for all developed countries. In most urbanised areas space is a scarce commodity. Therefore, a better management of the existing roads to increase or maintain their capacity level is the only solution. Research in the last two decades has focused on Intelligent Transport Systems (ITS) development. Predicting traffic flow in real or quasi-real-time can be used to prevent or alleviate future congestion [1]. The key to an effective proactive method is a model that produces timely and accurate predictions for not only recurrent traffic congestion (normal) but also non-recurrent (abnormal) congestion.

An urbanised arterial road is a major high capacity and large traffic volume road located in a built-up area. Urbanised arterial roads have limited entry and exit points and are used as thoroughfares for travelling substantial distances between major geographical locations, such as motorways or city centres.

Therefore, they are often used for commuting and the transportation of goods. The importance of keeping these roads free of congestion is imperative, as severe congestion can result in traffic delays extending into various other traffic networks. Due to a high volume of heterogeneous traffic, arterial roads tend to be consistently busy and contain different seasonality patterns across different vehicle classes when compared with motorways or city centres. Additionally, due to operating at close to capacity levels, their traffic flow tends to be more volatile. Therefore, research into traffic prediction models for this class of road is vital. One key component of arterial roads which could improve the prediction is the latent patterns embedded within the different vehicle classes. However, there is a lack of research on arterial roads, particularly using vehicle classes as a variable. Machine learning models are a popular option for traffic prediction [2], which are capable of learning complex and latent patterns embedded in volatile traffic data containing both normal and abnormal traffic congestion.

The contributions of this work include: 1) we have examined and compared the traffic flow prediction accuracy and time-horizon sensitivity of existing machine learning models for urbanised arterial roads, based on a real dataset; and 2) we have investigated different input parameter settings (different classes of vehicles) to understand how heterogeneous traffic flow affects the machine learning models' prediction accuracy for urbanised arterial roads.

The rest of this paper is organised as follows: Section II presents the State of the Art Machine Learning Traffic Flow Prediction Models; Section III describes the Methodology used for the experimentation; Section IV details the Experimental Evaluation; and Section V discusses the Conclusion and Future Work.

II. STATE OF THE ART MACHINE LEARNING TRAFFIC FLOW PREDICTION MODELS

In this section we will focus on the three most popular machine learning algorithms used in road traffic prediction [2] and assess their suitability

for volatile traffic flow on urbanised arterial roads.

1) *K Nearest Neighbour Based Traffic Prediction Models:*

K Nearest Neighbour (KNN) [3] is a non-linear, non-parametric, lazy learning pattern recognition model, which is used predominantly for classification as well as regression. The model does not rely on periodicity in historical traffic data. Therefore, it is able to adapt to dynamic changes in road traffic data [4] [3] and predict volatile traffic flow. Furthermore, the model's simplicity allows it to be a computationally efficient model.

The most commonly used distance function in KNN road traffic prediction models is the Euclidean Distance Function (Equation 1).

$$\sqrt{\sum_{i=1}^k (x_t - y_t)^2} \quad (1)$$

Cai et al. in 2016 [5] used a four week dataset (weekdays only) containing traffic speed and flow with a time horizon of five minutes. The data was divided into fifteen days for training and five days for testing, to develop a spatio-temporal correlation KNN model. In traditional time-series KNN models, the data is marked by the time. However, Cai et al. used the physical distance of the road segments in a spatio-temporal matrix in the calculation. The distance function used was an Euclidean distance function (Equation 1). The results showed that the addition of a spatial parameter improved the accuracy of the prediction for single and multi time-steps. However, taking the data into a spatio-temporal matrix increases its dimension, thus, increasing its computational time and complexity. Furthermore, Cai used the Euclidean distance function stating it was the most commonly used for time series prediction. For traffic data which contains different types of variables (speed, distance, flow, etc) Cityblock/Manhattan Distance Function (Equation 2) would be most suitable.

$$\sum_{i=1}^k |x_t - y_t| \quad (2)$$

The most commonly used weights in KNN models is the inverse distance or attributes weights (Equation 3).

$$\hat{x} = \frac{\sum_{i=1}^k x_{i,t+h} w_i}{\sum_{i=1}^k w_i} \quad (3)$$

Where $\hat{x}$ is the predicted value, x_i is the neighbour from the training set, t is the time parameter, h is the time horizon, and w is defined as $\frac{1}{d_i}$, x_i is the neighbour, and d is the distance function. This allows the model to give priority to data that is closest in the feature space to the test data case, be that temporal, spatial, or other parameters such as traffic flow or weather. Researchers have sought to improve upon this simplistic method. Sun et al. in 2017 [1] proposed a model that weighted the k nearest neighbours dynamically based on the traffic flow rate known as the 'weighted parameter tuples' (WPT). The research used one year of highway data from Kunshi, Beijing, split 80% for training and 20% for testing,

at five minute time horizons. It concluded that dynamically weighted k values outperformed statically weighted k values. However, the model was never compared to other machine learning models such as Kalman Filters or Artificial Neural Networks. The model was also not tested on other road types.

In conclusion, KNN model is a well established non-parametric prediction model that requires no fitting to the road network and is able to handle and predict recurrent and non-recurrent traffic congestions.

2) *Support Vector Regression Based Traffic Prediction Models:*

Support Vector Regression (SVR) [6], developed by Scholkopf et al. in 1998, is an extension of a classifier model, Support Vector Machine (SVM) [7]. By including an extra slack variable for each training point, the model is able to predict continuous ordered variables (Equation 4 to 6).

$$f(x) = \langle w + \Phi x \rangle + b \quad (4)$$

Where $\langle w + \Phi x \rangle$ denotes the dot product of the original feature space (X), w is a weight vector in the higher dimensional feature space (F), Φ a mapping function, and b is the bias of the regression function.

$$\frac{1}{2} \|w\|^2 \quad \text{subject to} \quad \begin{aligned} y_i - \langle w, x_i \rangle - b &\geq \epsilon \\ \langle w, x_i \rangle + b - y_i &\geq \epsilon \end{aligned} \quad (5)$$

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\zeta_i + \zeta_i^*) \quad \text{subject to} \quad \begin{aligned} y_i - \langle w, x_i \rangle - b &\geq \epsilon + \zeta_i \\ \langle w, x_i \rangle + b - y_i &\geq \epsilon + \zeta_i^* \\ \zeta_i, \zeta_i^* &\geq 0 \end{aligned} \quad (6)$$

Where C is a constant, subject to $C > 0$, that determines the amount of deviations larger than ϵ that will be tolerated, and n is the number of data points.

A common adjustment to the SVR model is the replacement of the dot product ($\langle w, x_i \rangle$) in Equation 4 with a kernel function. This negates the need for the mapping function (Φ). In road traffic prediction models the most commonly used kernel function is the Radial Basis Function (RBF) (Equation 7). This is due to its highly effective mapping of non-linear data [8] into a higher dimensional feature space. It's a 'local' kernel function, therefore, it has a strong learning ability but weak generalisation. Thus, it is ideal for volatile traffic data. Extreme historical variances in the data will not negatively affect the future predictions.

$$k(x, y) = \exp\left(\frac{-\|x - y\|^2}{2\sigma^2}\right) \quad (7)$$

Hu et al. in 2016 [9] used three months of motorway data with a time horizon of fifteen minutes to develop a SVR and Particle Swarm Optimisation (PSO) hybrid. The research focused on replacing the standard equation (Equation 6), which calculates the optimal parameters for the SVR, with a PSO. This was done to reduce the time taken during the learning step and aid real-time prediction. The results were compared to various other ML algorithms such as KNN and ANN. The results were comparable but not significantly close. However,

the study failed to compare the computation time of the model against other ML models, which appeared to be the premise of the research. Only times for the PSO-SVR variations were compared. The study also failed to take into account any spatial parameters. It only considered traffic speed, flow and journey time.

Ling et al. in 2017 [10] improved upon the Hu et al.'s work further by using a PSO that not only took into account historical data but also real-time data; an adaptive particle swarm optimisation (APSO). It also sought to improve the mapping function. As road traffic data is both random and non-linear [11], a 'multi-kernel' approach was developed. A hybrid of an RBF and a polynomial kernel (Equation 8) was used.

$$k(x, y) = xy^t \quad (8)$$

This was deemed more suitable for volatile road traffic data as it is able to utilise the RBF's ability for local learning and the polynomial's ability for global generalization. The results for the RBF, polynomial and hybrid kernel were Mean Absolute Percentage Error (MAPE) of 15.76, 16.52, and 10.57 respectively. The study was tested on both urban and freeway roads. However, although the study did use volatile rush hour data, no abnormal traffic data was used. In theory, a polynomial kernel should negatively affect the model under these conditions. In conclusion, although a SVR model is at its core a linear classifier, it is still able to cope with and predict non-linear data. However, its choice of kernel is key.

3) Artificial Neural Networks Based Traffic Prediction Models:

Artificial Neural Networks (ANN) [12] is a non-parametric flexible prediction method that can model the complex latent non-linear relationships between the input and output data [13] without any prior knowledge of the parameters. Therefore, ANN have been applied to road traffic prediction due to their ability to handle and predict non-linear volatile data.

Although Back Propagation Neural Networks (BPNN) can produce good prediction results, they can be improved by using different training procedures, different internal structures/mathematics, preprocessing of the input data, or including spatial and/or temporal patterns in the model [14]. More et Al. in 2016 [15] developed a model to improve the internal structure by using a Jordan Sequential Neural Network (JSNN), a type of recurrent neural network (RNN). An ANN structure is based on latent patterns present in the historical data. However, these patterns may vary to those in the test data as traffic flow is complex and stochastic [16]. This leads to inaccuracies in the prediction. To overcome this the JSNN based model adds 'context' to the prediction by allowing it to 'remember' the most recent past data (partially recurrent network). As shown in Fig. 1, the hidden layer receives input from the current time-step (t) and context from the context layer from the previous time-step ($t - 1$). The JSNN model worked with "almost 98% accuracy for most of the datasets" [15]. However, the study failed to provide any standard performance measures. The model was also not

compared to any other ANN models to assess its success. Goves et al. in 2016 [17] took a different approach. Majority of

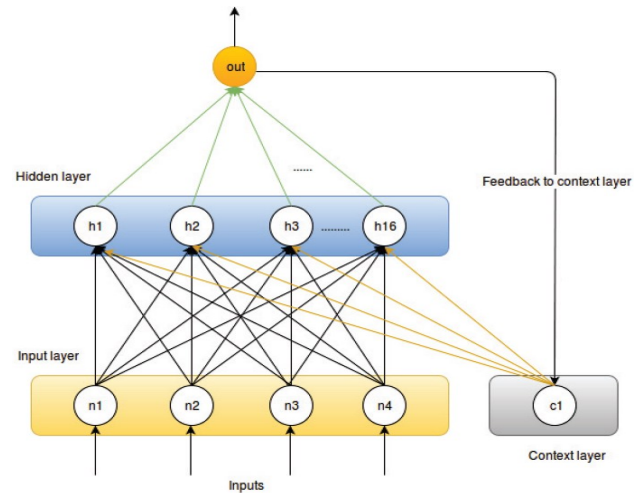


Fig. 1. Jordan Sequential Neural Network [15]

research considered only one geographical site or sensor [17], whereas Goves et al. developed an ANN that made predictions for a network of sensors. This was done by improving the preprocessing of the input data. Training data of 102 traffic detectors spread over 20km of roads on the M60, M62 and M602 near Manchester, UK, with a time horizon of fifteen minutes was used. The data was compressed in three ways. Firstly, the model only considered the density parameter ($\frac{volume}{speed}$), thus, negating the need for speed and volume parameters. Secondly, the data across three lanes of a road was aggregated into one average value. Lastly, the dimensions of the data were then reduced through the use of an auto-encoder. Once the data had been compressed it was standardised and put through three different BPNNs. They contained one, two or three hidden layers. The predictions were then decompressed using the auto-encoder and compared. The results showed a trade-off between dimensionality reduction and accuracy. Nevertheless, the predictions obtained were accurate to within two vehicles, per kilometre, per traffic lane, 90% of the time. However, the study only predicts vehicle density. This is a helpful indicator of the traffic status but traffic planners and road users would need to know the speed and volume. For example, is density made up of few cars moving very slowly or lots of cars moving very fast? Also, aggregating the lanes presents a problem. An issue can be identified but not known if it is all lanes or isolated to one. In conclusion a RNN can handle and predict volatile traffic data. Furthermore, due to the nature of neural networks, it does not need to find the complex latent patterns in all input parameters. Instead it's able to approximate them well given a single input parameter and an output through deep architecture, allowing input data to be simplified, thus, improving computational times.

The above models mainly focused on one geographical location, such as a motorway, and normal traffic data. Little

research has been done on urbanised arterial roads using abnormal traffic data. This may be due to traffic flow being more complex than other road layouts due to their interconnected nature and composition of vehicles classes, which varies from geographical location to time of day. Therefore, ratio of cars to trucks can fluctuate with both classes containing their own latent seasonal patterns. Research into urbanised arterial roads using different vehicle classes as a dynamic input parameter could improve prediction accuracy. However, research using truck flow as an input parameter is also limited [2]. Therefore, in this work we investigate how these machine learning models can be applied to traffic prediction on urbanised arterial roads.

III. METHODOLOGY

In this section, we are mainly focusing on two goals: 1) comparing the performance of existing machine learning models to determine which is superior to use for urbanised arterial roads between Manchester and Liverpool; and 2) determining if the use of vehicle classes as separate input parameters can improve the prediction of road traffic congestion on urbanised arterial roads with heterogeneous traffic flow.

To achieve the first goal, two experiments were carried out. Firstly, we have applied three machine learning models (KNN, SVR, and ANN) to a real dataset and compared the accuracy of their predictions. Furthermore, the second experiment compared the models' sensitivity to time-horizon changes. The prediction time horizon was changed from five to twenty minutes in five minutes intervals, and the models' prediction accuracy was compared to determine which produced the most accurate prediction. This was done as it was noted that there is no standard time-horizon for short-term traffic prediction. Publications range from one to ninety minutes, with the most commonly used being five.

For the second goal, to the best of our knowledge no research has investigated the effect of all vehicle classes on the accuracy of road traffic prediction. Therefore, two experiments were carried out. Firstly, an experiment was performed to compare the prediction of traffic flow using only the total vehicles and the individual vehicle class totals. In addition, a second experiment was conducted comparing the predictions accuracy using each vehicle class to determine which class or classes were most influential for prediction. The details of all experiments conducted using Matlab are detailed below.

1) KNN Model:

During the experiments the distance function used was the Euclidean distance function (Equation 1). This was due to all input parameters being of the same type (number of vehicles). The value of k was optimised by comparing the prediction values for k between 30 and 90, in increments of 5. The k that produced the optimal prediction based on the lowest empirical error was used and compared to other models. The prediction function used was the inverse distance weighting (IDW) function (Equation 3). This function gives priority to closer neighbours than those further away, thus, making it more reliable for predicting extreme traffic conditions (if these conditions have been seen before).

2) SVR Model:

In the SVR model the dot product in Equation 4 was replaced with the RBF kernel function (Equation 7), the most commonly used function for road traffic predictions. This was due to its ability to map non-linear data [8]. It has strong learning ability but weak generalisation, allowing the model to better handle volatile traffic data, as extreme historical variances will not negatively affect the future predictions. Additionally, Cherkassky et al. in 2004 [18] investigated the selection of hyper parameters for SVR models. They concluded that the parameter selection of c , which balances the models complexity and estimation error, and ϵ , which is used to fit the training data, can be derived directly from the training data. Therefore, the values of c and ϵ used in the experiments can be found using Equations 9 and 10 [18].

$$c = \max(|\hat{y} + 3\sigma_y|, |\hat{y} - 3\sigma_y|) \quad (9)$$

$$\epsilon \in [0, 0.1] \quad (10)$$

3) ANN Model:

The ANN model used in the experiment was an open loop time-series non-linear autoregressive network with exogenous inputs (NARX). NARX is a popular ANN model used in time-series prediction. It is able to predict traffic flows efficiently as it bases its predictions on historical data, feedback (recurrent), and exogenous input (current and historical data). Therefore, it is able to 'remember' the recent past and handle volatile data [15]. The transfer function used was a Sigmoid Transfer Function (Equation 11). This is a standard transfer function used for time-series data.

$$S(x) = \frac{e^x}{e^x + 1} \quad (11)$$

A. Dataset Description

The data used for the experiments was a real-life dataset of an urbanised arterial road. It contains three months of traffic flow volumes, broken down into vehicle classes, as shown in Table I, and contains a time horizon of five minutes. Therefore, the input parameter used in the experiments were the different vehicle classes. This data was used due to being a typical urbanised arterial road. It has a heterogeneous traffic flow, containing high proportion of trucks and HGVs typically found on these road types. Furthermore, the data contains both normal and abnormal traffic data. For experimental purposes

TABLE I
CATEGORIES OF VEHICLE TYPE

Class No.	Vehicle Type
1	Motorcycles
2	Car or Van
3	Car or Van with Trailer
4	Rigid Goods
5	Articulated HGV
6	Bus or Coach

the data was split. Two months of data was used to train the models and the remaining one month was used for testing purposes.

B. Performance Metrics

In order to compare the models a performance metric was used. A standard performance metric, Root Mean Squared Error (RMSE) was used (Equation 12).

$$RMSE = \sqrt{\frac{\sum_{t=1}^n (F_t - A_t)^2}{n}} \quad (12)$$

where F_t is the predicted value at time t , A is the actual value at time t , and n is the number of time steps predicted.

IV. EXPERIMENTAL EVALUATION

1) Comparison of Existing Machine Learning Models:

To achieve goal one the following two experiments were performed:

- Experiment One: The machine learning models were compared on their empirical prediction accuracy, using the same real dataset, to determine which model performed the best.
- Experiment Two: The machine learning models were compared on their empirical accuracy, using various time-horizons (five to twenty minutes in five minute intervals) and the same real dataset, to determine the models time-horizon sensitivity.

Experiment One

Table II shows the set-up parameters used during each run of Experiment One. It should be noted that the value of k in the KNN model was 65, the value that produced the lowest empirical root-squared mean error (RSME), in all experiments.

TABLE II
PARAMETER SET-UP FOR EXPERIMENT ONE

Parameter	Details
Time Horizon	5 Minutes
Training Data Size	2 Months
Test Data Size	1 Month
Input Parameters	All Vehicle Class Totals
Parameter Changed	Model Only

Experiment One Results

In Figures 2, 4, and 6, the red lines represent the predicted and blue lines the actual traffic flow values. The y axis is total number of vehicles, which is plotted against the x axis, time. Experiment One's results (Table III) show that the most successful traffic prediction model for the urbanised arterial road between Manchester and Liverpool was the ANN model, with a RMSE of 16.56% compared to the KNN and SVR which had a RMSE of 19.35% and 19.79% respectively.

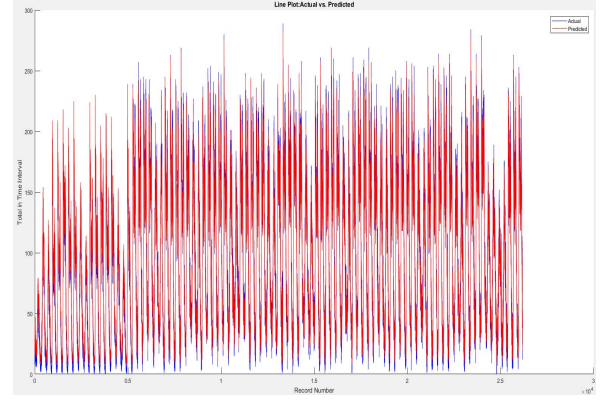


Fig. 2. KNN Model's Actual and Predicted Traffic Flow Plotted Against Time

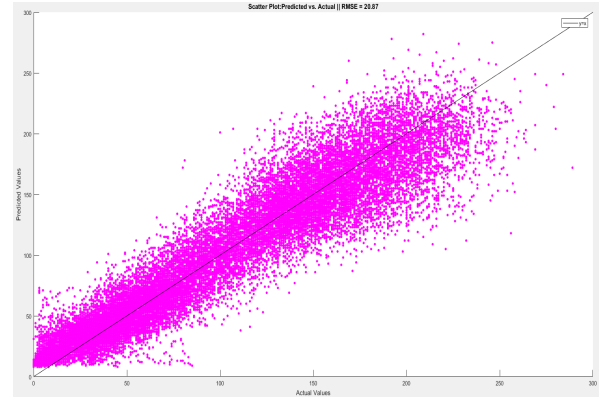


Fig. 3. KNN Model's Actual vs Predicted Traffic Flow

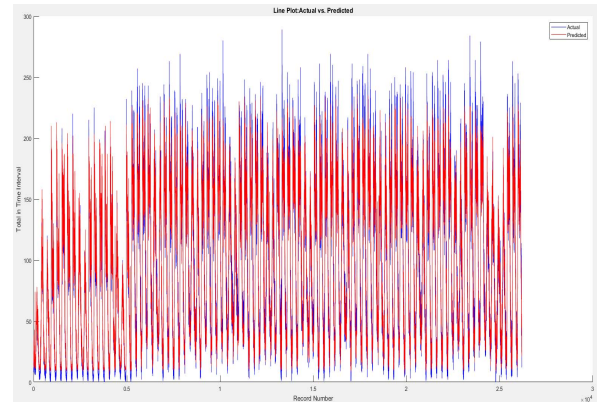


Fig. 4. SVR Model's Actual and Predicted Traffic Flow Plotted Against Time

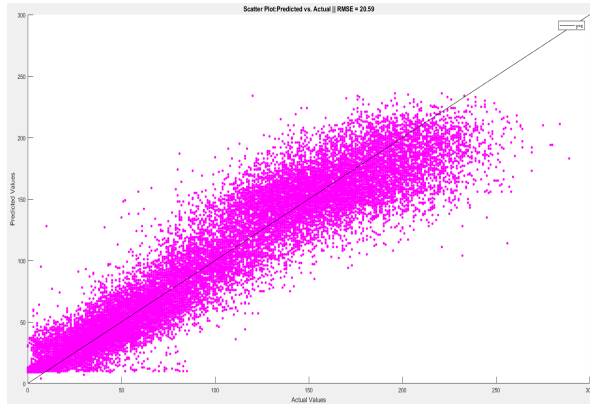


Fig. 5. SVR Model's Actual vs Predicted Traffic Flow

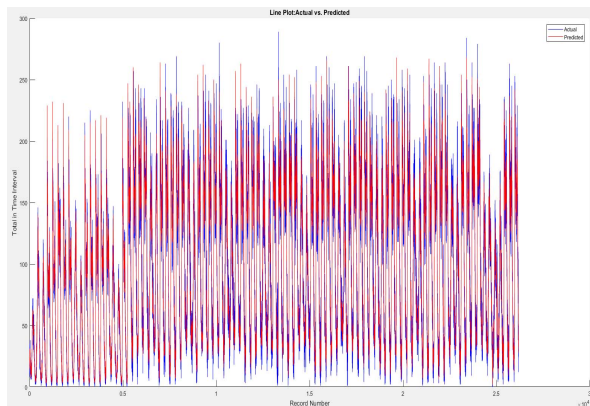


Fig. 6. ANN Model's Actual and Predicted Traffic Flow Plotted Against Time

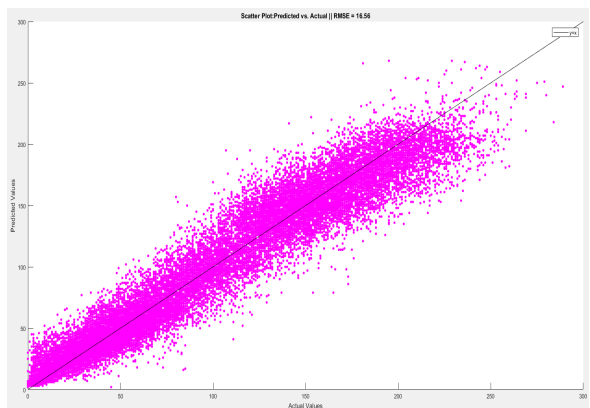


Fig. 7. ANN Model's Actual Predicted Traffic Flow

TABLE III
THE RMSE OF EACH MODEL

Model	RMSE (%)
KNN	19.35
SVR	19.79
ANN	16.56

Experiment Two

Table IV shows the set-up parameters used during each run of Experiment Two.

TABLE IV
PARAMETER SET-UP FOR EXPERIMENT TWO

Parameter	Details
Time Horizon	5 to 20 Minutes
Training Data Size	2 Months
Test Data Size	1 Month
Input Parameters	All Vehicle Class Totals
Parameter Changed	Model & Time Horizon

Experiment Two Results

Table V and Fig 8 presents the RMSE (in percentage form) for all models and time horizons. The results show that the most successful time horizon was the shortest, five minutes. Although not explicitly stated in publications, this may be the rationale behind why five minutes is the most commonly used. ANN, again, was the most successful prediction, however, it was also the most sensitive to, with the RMSE increasing by 4.64%. KNN and SVR in comparison appear to fluctuate with the general trend, based on the limited data, shows that the prediction accuracy progressively declines.

TABLE V
THE TIME-HORIZON SENSITIVITY OF EACH MODELS MEASURED BY THE RMSE

Model	5 Minutes	10 Minutes	15 Minutes	20 Minutes
KNN	19.35	18.68	19.60	19.10
SVR	19.79	20.37	21.50	21.42
ANN	16.56	17.25	18.54	21.20

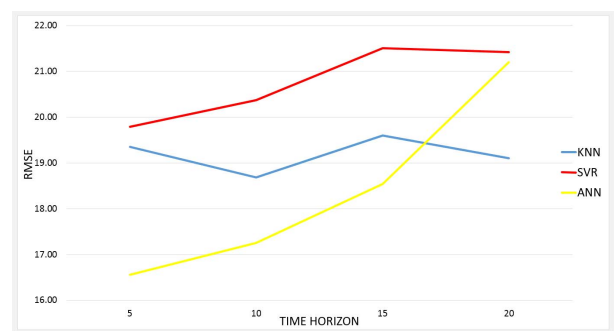


Fig. 8. The Time Horizon Sensitivity of Each Model Shown by RSME vs the Time Horizon

2) The Effect of Heterogeneous Traffic Flow on Prediction Accuracy:

To achieve goal two, determine if the use of vehicle classes as separated input parameters can improve the prediction of road traffic congestion on urbanised arterial roads between Manchester and Liverpool, using real-life data, the following two experiments were performed:

- Experiment Three: The machine learning models were compared using the same real dataset, while changing the input parameters from total traffic flow to all individual vehicle classes total traffic flow. This was done to determine if the latent patterns of each vehicle class would affect the prediction accuracy.
- Experiment Four: The machine learning models were compared using the same real dataset, while using one vehicle class total as an input parameter. This was done to determine which vehicles classes were most influential for predicting traffic congestion.

Experiment Three

Table VI shows the set-up parameters used during each run of experiment three. Each model was tested and compared using a different input parameters. The input parameters used were the total vehicle traffic flow (a singular input parameter) or the individual vehicles class totals (six input parameters).

TABLE VI
PARAMETER SET-UP FOR EXPERIMENT THREE

Parameter	Details
Time Horizon	5 Minutes
Training Data Size	2 Months
Test Data Size	1 Month
Input Parameters	All Vehicle Class and Total Traffic Flow
Parameter Changed	Model & Vehicle Classes Total

Experiment Three Results

Table VII shows that, for all models, using the individual vehicle class totals as an input parameter improves traffic flow prediction when compared to predictions based only on total traffic flow. Therefore, taking into consideration the latent patterns for each vehicle class does improve prediction accuracy.

TABLE VII
COMPARISON OF THE MODELS' RMSE (%) BASED ON THE INPUT PARAMETERS OF TOTAL VEHICLE FLOW OR TOTAL OF ALL CLASSES OF VEHICLES

Model	Input of All Vehicle Classes	Input of Total No. of Vehicles
KNN	19.35	20.01
SVR	19.79	19.89
ANN	16.56	16.95

Experiment Four

Table VIII shows the set-up parameters used during each run of experiment four. Each model was tested and compared using a different singular input parameter. The input parameters used were the total of all classes, Class One, Class Two, Class Three, Class Four, Class Five, or Class Six of vehicle traffic

flow. It should be noted that the value of k in the KNN model was 65, the value that produced the lowest empirical root-squared mean error (RSME).

TABLE VIII
PARAMETERS FOR EXPERIMENT FOUR

Parameter	Detail
Time Horizon	5 Minutes
Training Data Size	2 Months
Test Data Size	1 Month
Input Parameters	One Vehicle Class
Variable Changed	Model & Individual Vehicle Class

Experiment Four Results

Based on Experiment Four's results, Class Two (car or van) is the most influential vehicle class for road traffic prediction, followed by Class Four (rigid goods) for all models. It is also worth noting that the results are better when only considering Class Two vehicle total, as oppose to all vehicle class totals.

TABLE IX
DIFFERENT CLASSES OF VEHICLES USED AS INPUT PARAMETERS

Model	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
KNN	19.96	18.60	20.01	19.92	20.01	20.01
SVR	19.94	18.53	19.99	19.90	20.01	20.00
ANN	16.84	16.40	16.92	16.83	16.93	16.85

V. CONCLUSION

This study conducted a comparative analysis of three machine learning algorithms (KNN, SVR, and ANN). The three machine learning algorithms were applied to a real data set to determine: 1) which machine learning model produced the most accurate prediction and how time horizon changes affect their accuracy (time horizon sensitivity); and 2) the impact of vehicle classes as input parameters on prediction accuracy. The experimental results presented show that the most accurate machine learning model was the ANN. It consistently produced the most accurate prediction. However, the ANN model was the most sensitive to time horizon changes, as seen in Table V, but still produced the most accurate prediction in all time horizons tested. Therefore, it can be concluded that ANN models are suitable for short-term traffic flow prediction on urbanised arterial roads. ANN should be explored further for urbanised arterial roads and heterogeneous traffic flow. The experimental results also show that using the individual vehicle class totals, as opposed to the total number of vehicles, improves all three machine learning models' predictions (as seen in Table VII). This is likely due to the models being able to ascertain latent patterns within the different vehicle class data to improve prediction accuracy. The composition of vehicles on urbanised arterial roads varies from geographical area to time of day. The ratio of cars to trucks can fluctuate with both classes having their own latent seasonal patterns. Therefore, considering each class as a separate input parameter has improved the prediction accuracy. In addition, the most influential vehicle classes were Class Two (car or van) and Class Four (Rigid Goods). This may be due to these classes

being the main classes in terms of volume in the data. However, it is worth noting that prediction accuracy improved when considering only Class Two as an input parameter compared with using all vehicle class totals. This suggests that the latent seasonal pattern within the main vehicle class type (Class Two in this research) is key to the prediction of road traffic flow. This should be researched further to determine if there is a dynamic optimal use or weighting of vehicle classes that could improve prediction accuracy further.

Based on the experimental results in this paper future research on the prediction of road traffic flow on urbanised arterial roads should focus the internal structure of ANN, especially ANNs with deep architecture.

ANNs with deep architectures should be explored further. A deeper architecture could enable the ANN to obtain a more complex, non-linear function and thus, improve prediction accuracy. Although, the deep architecture and computing cost must be balanced to ensure the model has real-life application. In addition, more datasets will be collected and used for validation of the models.

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REFERENCES

- [1] Bin Sun, Wei Cheng, P. Goswami, and Guohua Bai, "Flow-aware WPT k-nearest neighbours regression for short-term traffic prediction," in *2017 IEEE Symposium on Computers and Communications (ISCC)*. IEEE, 7 2017, pp. 48–53. [Online]. Available: <http://ieeexplore.ieee.org/document/8024503/>
- [2] S. Suhas, V. V. Kalyan, M. Katti, B. V. A. Prakash, and C. Naveena, "A Comprehensive Review on Traffic Prediction for Intelligent Transport System," in *2017 International Conference on Recent Advances in Electronics and Communication Technology (ICRAECT)*. IEEE, 3 2017, pp. 138–143. [Online]. Available: <http://ieeexplore.ieee.org/document/8081917/>
- [3] G. A. Davis and N. L. Nihan, "Nonparametric Regression and Short-Term Freeway Traffic Forecasting," *Journal of Transportation Engineering*, vol. 117, no. 2, pp. 178–188, 3 1991. [Online]. Available: <http://ascelibrary.org/doi/10.1061/%28ASCE%290733-947X%281991%29117%3A2%28178%29>
- [4] B. M. Williams and L. A. Hoel, "Modeling and Forecasting Vehicular Traffic Flow as a Seasonal ARIMA Process: Theoretical Basis and Empirical Results," *Journal of Transportation Engineering*, vol. 129, no. 6, pp. 664–672, 11 2003. [Online]. Available: <http://ascelibrary.org/doi/10.1061/%28ASCE%290733-947X%282003%29129%3A6%28664%29>
- [5] P. Cai, Y. Wang, G. Lu, P. Chen, C. Ding, and J. Sun, "A spatiotemporal correlative k-nearest neighbor model for short-term traffic multistep forecasting," *Transportation Research Part C: Emerging Technologies*, vol. 62, pp. 21–34, 1 2016. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0968090X15003812>
- [6] B. Scholkopf, P. Simard, A. Smola, and V. Vapnik, "Prior Knowledge in Support Vector Kernels," 1998. [Online]. Available: <http://papers.nips.cc/paper/1457-prior-knowledge-in-support-vector-kernels.pdf>
- [7] K.-R. Muller, S. Mika, G. Ratsch, K. Tsuda, and B. Scholkopf, "An Introduction to Kernel-Based Learning Algorithms," *IEEE Transactions on Neural Networks*, vol. 12, no. 2, pp. 181–201, 3 2001. [Online]. Available: <http://ieeexplore.ieee.org/document/914517/>
- [8] M. T. Asif, J. Dauwels, Chong Yang Goh, A. Oran, E. Fathi, Muye Xu, M. M. Dhanya, N. Mitrovic, and P. Jaillet, "Spatiotemporal Patterns in Large-Scale Traffic Speed Prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 15, no. 2, pp. 794–804, 4 2014. [Online]. Available: <http://ieeexplore.ieee.org/document/6678316/>
- [9] W. Hu, L. Yan, K. Liu, and H. Wang, "A Short-term Traffic Flow Forecasting Method Based on the Hybrid PSO-SVR," *Neural Processing Letters*, vol. 43, no. 1, pp. 155–172, 2 2016. [Online]. Available: <http://link.springer.com/10.1007/s11063-015-9409-6>
- [10] X. Ling, X. Feng, Z. Chen, Y. Xu, and H. Zheng, "Short-term traffic flow prediction with optimized Multi-kernel Support Vector Machine," in *2017 IEEE Congress on Evolutionary Computation (CEC)*. IEEE, 6 2017, pp. 294–300. [Online]. Available: <http://ieeexplore.ieee.org/document/7969326/>
- [11] L. Zhang, Q. Liu, W. Yang, N. Wei, and D. Dong, "An Improved K-nearest Neighbor Model for Short-term Traffic Flow Prediction," *Procedia - Social and Behavioral Sciences*, vol. 96, pp. 653–662, 11 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877042813022027>
- [12] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *The Bulletin of Mathematical Biophysics*, vol. 5, no. 4, pp. 115–133, 12 1943. [Online]. Available: <http://link.springer.com/10.1007/BF02478259>
- [13] G. Zhang, "Time series forecasting using a hybrid ARIMA and neural network model," *Neurocomputing*, vol. 50, pp. 159–175, 1 2003. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0925231201007020>
- [14] C. P. I. Van Hinsbergen, J. Lint, and F. M. Sanders, "Short Term Traffic Prediction Models," in *14th World Congress on Intelligent Transport Systems, ITS*, 2007. [Online]. Available: https://www.researchgate.net/publication/228882378_Short_Term_Traffic_Prediction_Models
- [15] R. More, A. Mugal, S. Rajgure, R. B. Adhao, and V. K. Pachghare, "Road traffic prediction and congestion control using Artificial Neural Networks," in *2016 International Conference on Computing, Analytics and Security Trends (CAST)*. IEEE, 12 2016, pp. 52–57. [Online]. Available: <http://ieeexplore.ieee.org/document/7914939/>
- [16] A. Koesdwiady, R. Soua, and F. Karay, "Improving Traffic Flow Prediction With Weather Information in Connected Cars: A Deep Learning Approach," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 12, pp. 9508–9517, 12 2016. [Online]. Available: <http://ieeexplore.ieee.org/document/7501574/>
- [17] C. Goves, R. North, R. Johnston, and G. Fletcher, "Short Term Traffic Prediction on the UK Motorway Network Using Neural Networks," *Transportation Research Procedia*, vol. 13, pp. 184–195, 1 2016. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S2352146516300199#>
- [18] V. Cherkassky and Y. Ma, "Practical selection of SVM parameters and noise estimation for SVM regression," *Neural networks : the official journal of the International Neural Network Society*, vol. 17, no. 1, pp. 113–26, 1 2004. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pubmed/14690712>